

1. List of companies conducting placement drives using DSA?

Tier-1	Tier-2	Indian
Google	Oracle	Flipkart
Microsoft	Intel	Phone Pe
Amazon	Qualcomm	Swiggy
Apple	Samsung	Zomato
Adobe	Paypal	Meesha
NVIDIA	Goldman	Groww
Uber	Sachs	CRED
LinkedIn	Morgan	Zoho
Meta	Stanley	Freshwork

Service - Based

Tata	Accenture
Infosys	Capgemini
Cognizant	HCL Tech
Wipro	ITI Mindtree

2. What is the syllabus of DSA in gate?

1. Programming & Data Structures

- => Programming in C
- => Functions
- => Recursion
- => Arrays
- => Strings
- => Linked Lists [Singly, Doubly, Circular]
- => Stacks
- => Queues [Simple, Circular, Priority]

=> Trees

(i) Binary Trees

(ii) Binary search Trees (BST)

=> Binary Heaps

=> Graphs

2. Algorithms

=> Asymptotic Notations

1. Big-O (O)

2. Omega (Ω)

3. Theta (Θ)

=> Time complexity Analysis

=> Space complexity Analysis

=> Searching Algorithms

1. Linear Search

2. Binary Search

=> Sorting Algorithms

1. Bubble Sort

2. Selection Sort

3. Insertion Sort

4. Merge Sort

5. Quick Sort

6. Heap Sort

=> Hashing

=> Algorithm Design Techniques

1. Divide and Conquer

2. Greedy Algorithms

3. Dynamic Programming

=> Graph Algorithms

1. Breadth - First Search (BFS)
2. Depth - First Search (DFS)
3. Minimum Spanning Tree (Prim's & Kruskal's)
4. Shortest path Algorithms (Dijkstra, Bellman - Ford basics)

3. Find out what are the set of variables in Amazon invoice bill

=> Order Information

1. Order ID
2. Order Place
3. Invoice Number
4. Invoice Date
5. Invoice Details Reference

=> Seller Information

1. Seller Name
2. Seller Address
3. GSTIN
4. PAN number
5. CIN number

=> Buyers Information

1. customer name
2. Billing Address
3. Shipping Address
4. State code
5. Place of Supply
6. Place of Delivery

=> Tax Information

1. Taxable value
2. GST
3. CGST
4. SGST
5. IGST
6. Tax Total

4. Find out set of variables and datatypes in gas Bill, current Bill.

Gas Bill:

variables

Bill - ID

customer ID

customer Name

Address

Provider Name

Meter Number

Billing period

Bill date

Due date

Current Meter Reading

gas consumed

unit rate

Fixed charges

gas charges

Tax

Discount

Late - Fee

Discount

Late - Fee

Total - Amount

Amount paid

Payment Status

Payment Method

Payment Date.

Data types

String

String

String

String

String

String

Date Range

date

date

Float

Float

Decimal

Decimal

Decimal

Decimal

Decimal

Decimal

Decimal

Decimal

String

String

Date.